

High-Performance Computing Workflow for Rapid-Response Physics-Based 3D Earthquake Simulations Integrated In Data Lake

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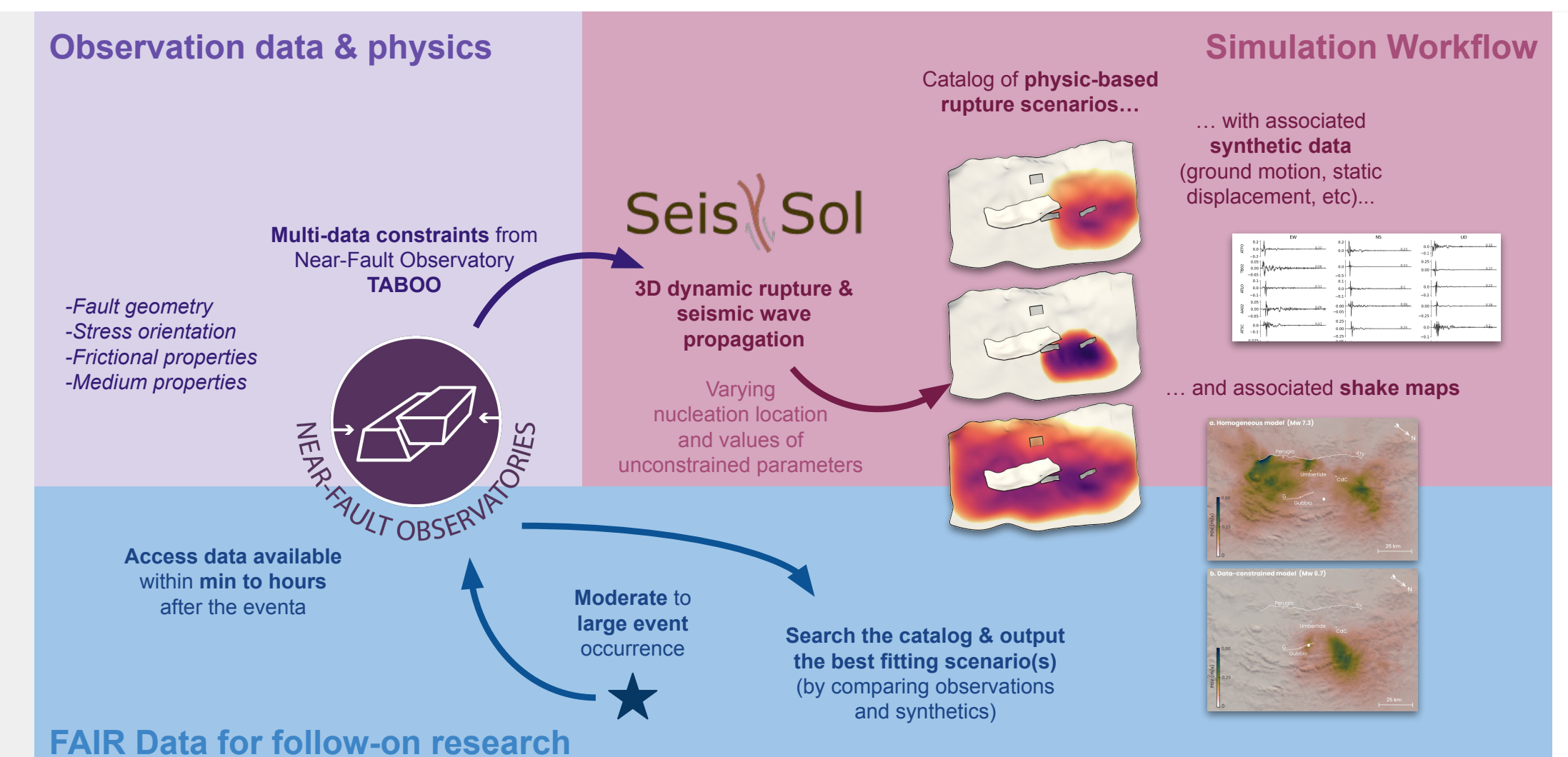


1. Motivation

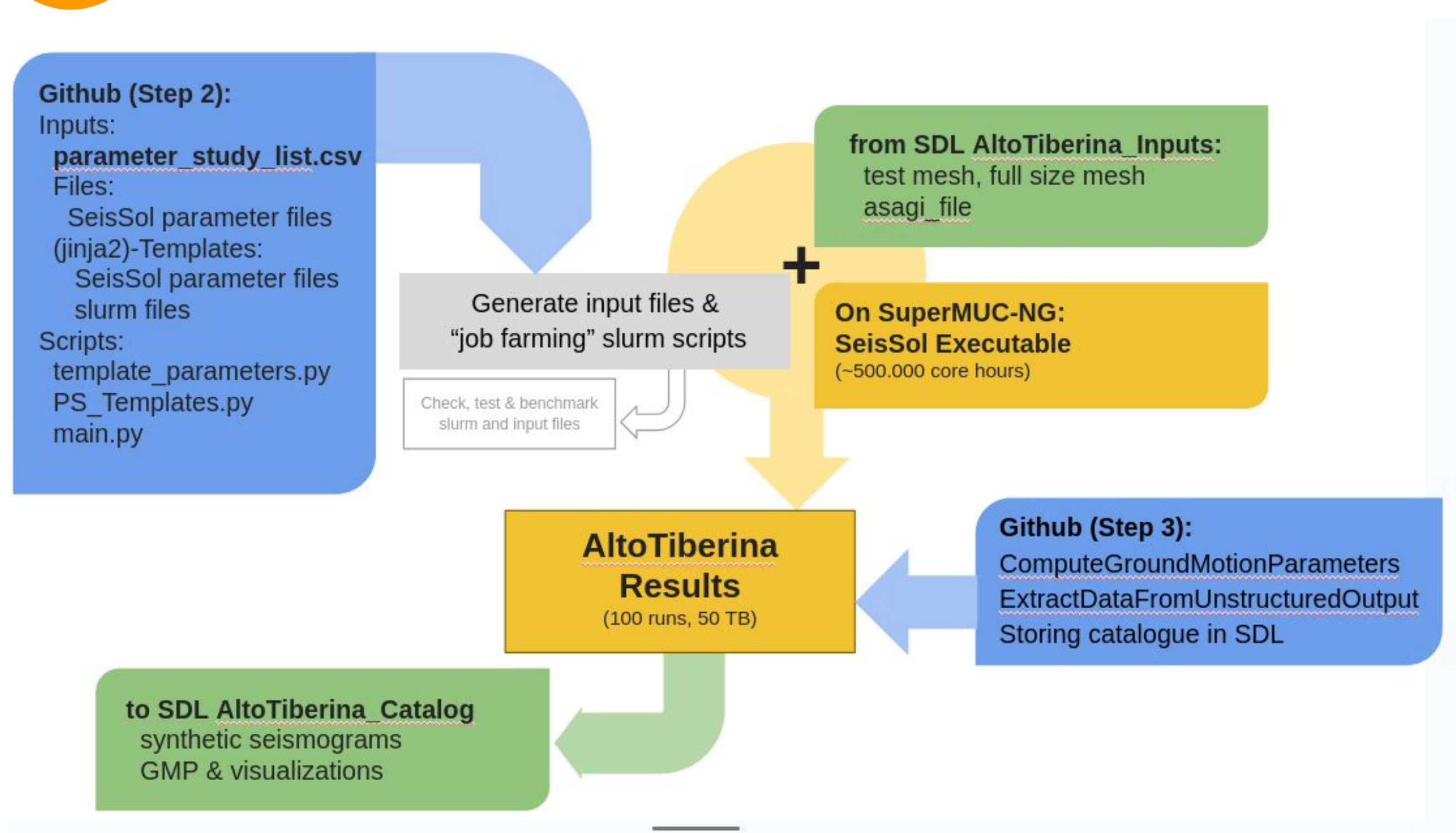
- Dynamic rupture simulations are the **gold standard** in earthquake (EQ) simulations
- Setting them up (retrieving data, meshing, calibration) after an EQ can take weeks
- Goal: rapid-response results within an hour after the EQ**

→ **Solution:** pre-computed EQ catalog and automated selection of closest fit

→ A new Simulation Data Lake is used for FAIR storage of the catalog



2. The Catalog Creation Workflow



Preparatory Work

Mathilde Marchandon et al. "Forecasting 3D Rupture Dynamics of the Alto Tiberina Low-Angle Normal Fault, Italy" In: *Seismica* 4.2 (2025), analyzed the fault to define the range of parameter values used for the catalog. [doi:10.26443/seismica.v4i2.1603](https://doi.org/10.26443/seismica.v4i2.1603)

4. The Rapid Response Workflow



Automatization

- Cronjob version for servers or desktop
- [Common Workflow Language](#) implementation and special-build docker image for cloud-enabled workflows (e.g. DT-Geo)

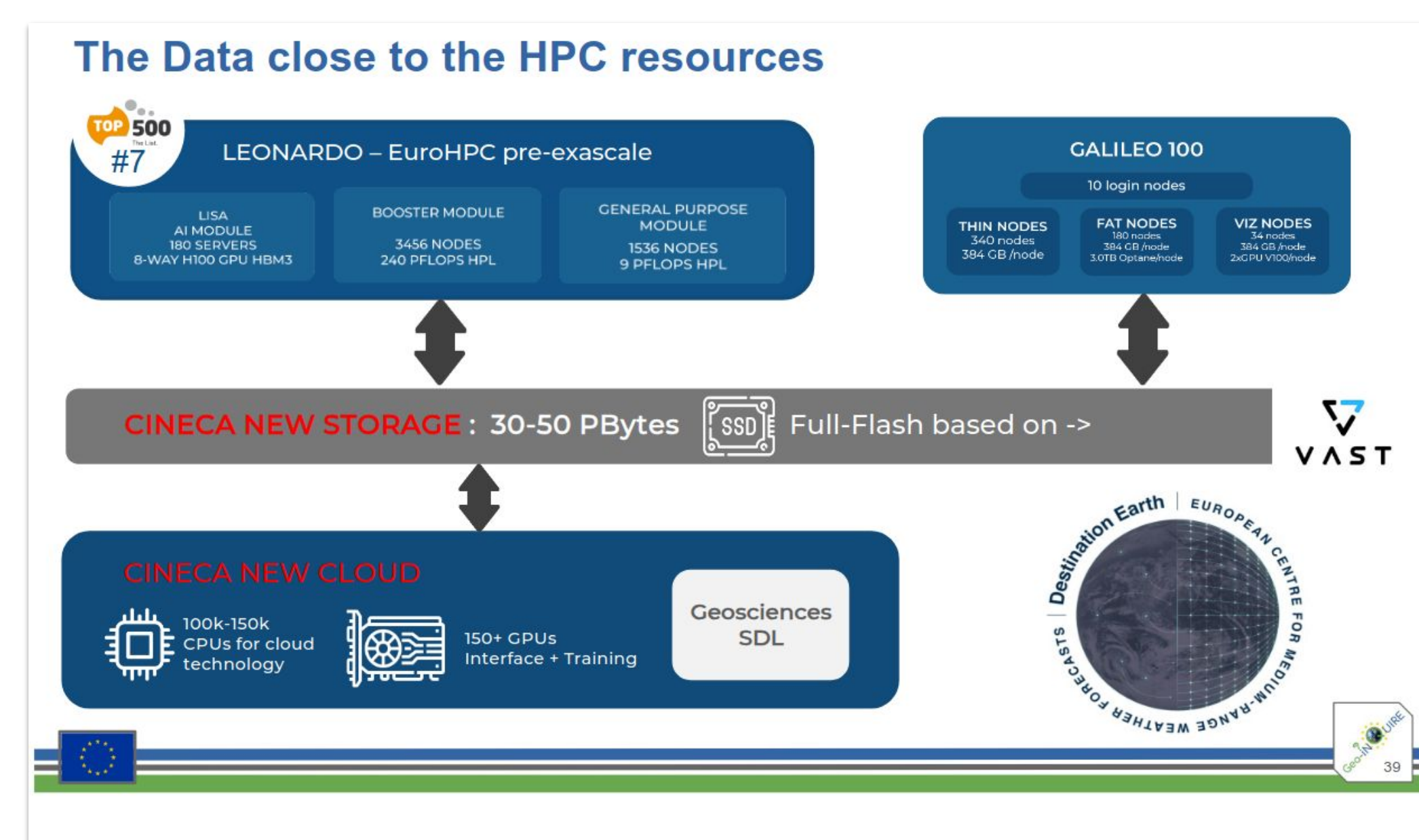
Key Takeaways

- Dynamic Rupture Earthquakes Simulations with SeisSol are the gold standard but take weeks after an event
- We need a first solution within an hour after the EQ so we pre-computed an catalog for Alto Tiberina
- For FAIR access the catalog is stored in the SDL
- The Rapid-Response Workflows is fully automated

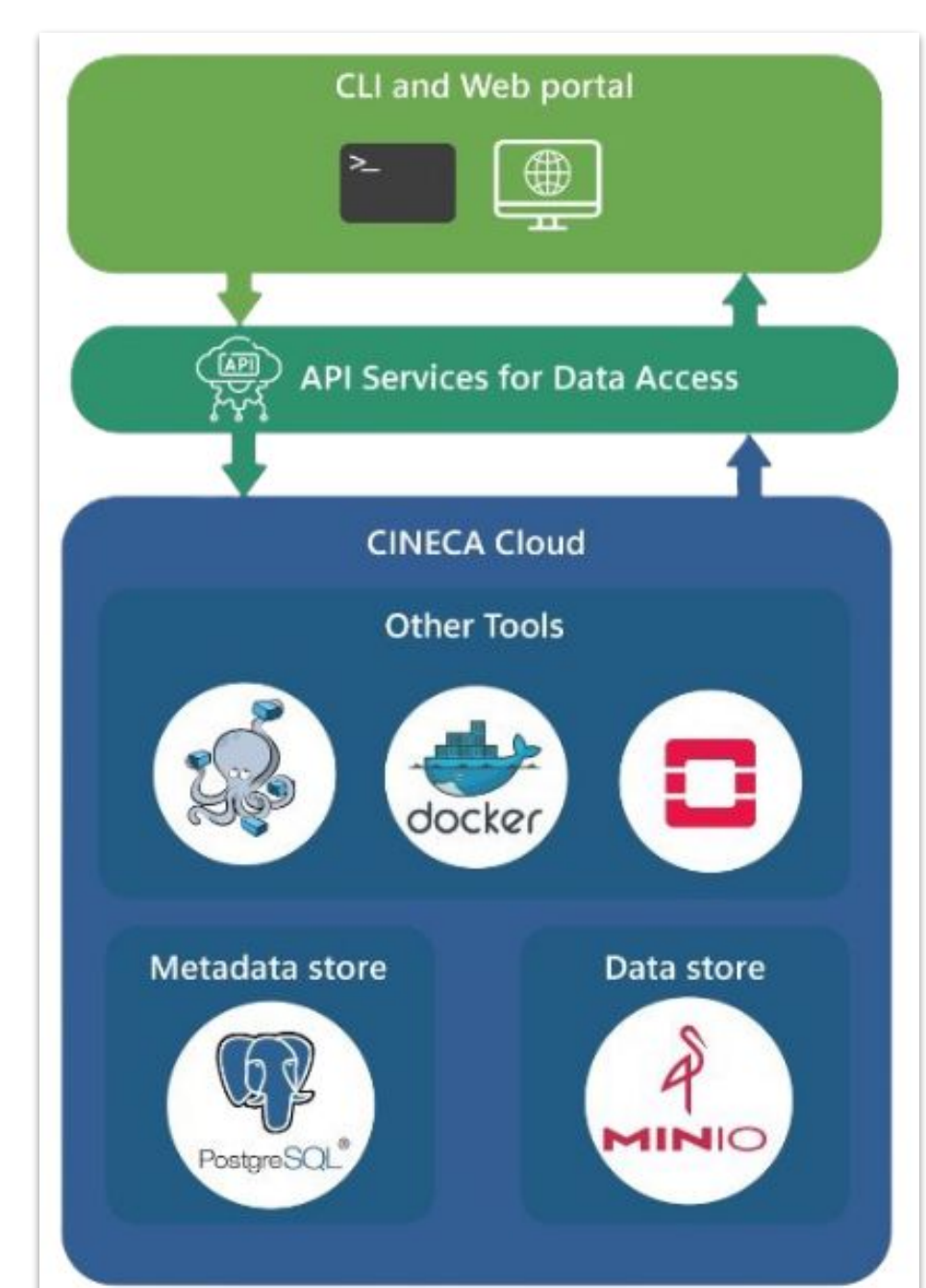
3. The Geo-INQUIRE Simulation Data Lake (SDL)

Data lake solutions are changing the Research Data Management (RDM) of HPC-intensive codes and parameter studies:

- FAIR (Findable, Accessible, Interoperable, Reusable) data sharing (metadata+DOIs)
- long-term storage of meshes, input and output datasets and products
- includes project results and DT-Geo Digital Twin Components (DTCs)
- accessible to all Earth Scientists in Europe and beyond
- HPC-close storage for fast access
- available to CINECA's AI-Factory



Hardware Overview of Geo-INQUIRE SDL ©CINECA



SDL Schematic overview ©CINECA

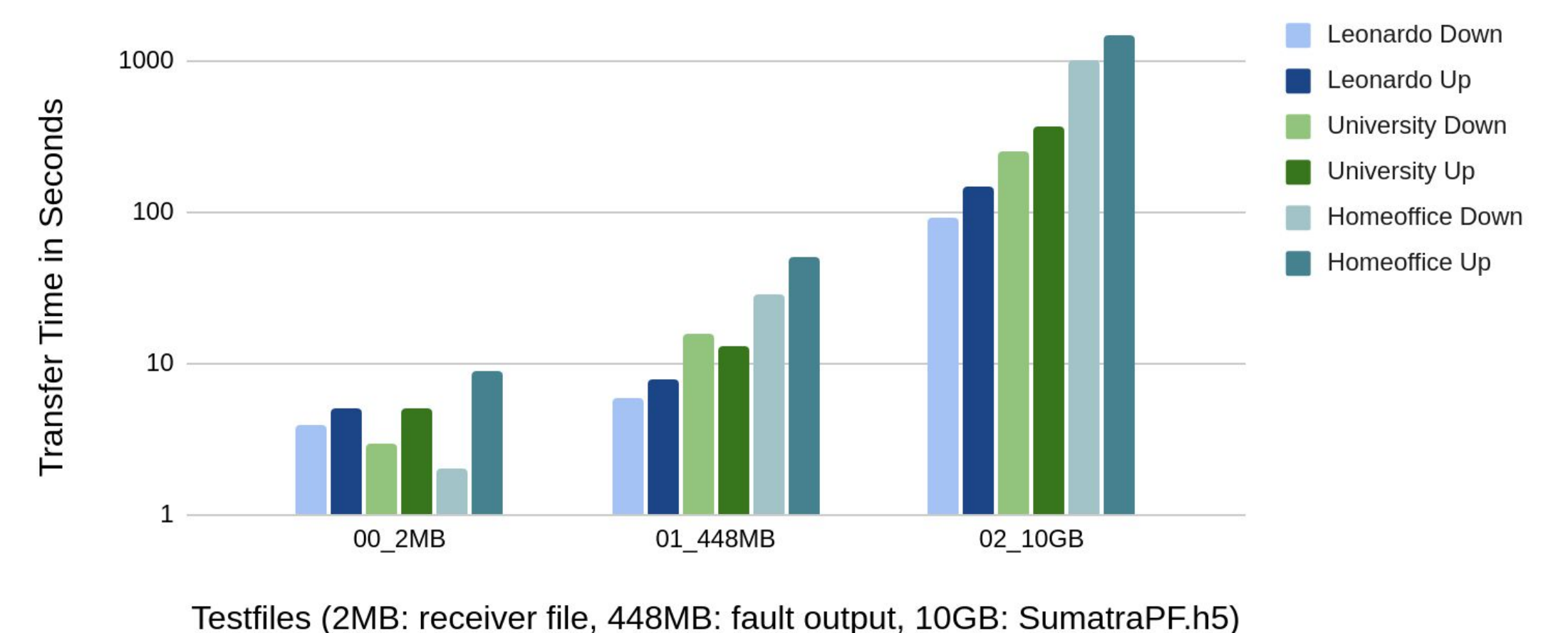
Benchmarking the Simulation Data Lake

We benchmarked the latest version of the SDL command line interface (v0.13.55) from Leonardo@CINECA, heisenbug@LMU and a laptop with a fibre optic connection.



SDL Benchmark

blue: Leonardo@CINECA, green: heisenbug@LMU, turquoise: Laptop@Home



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